

CHE 201 Fall 2019 Quiz 1

Marek Smirnow

TOTAL POINTS

75 / 100

QUESTION 1

1 Question 1 **40 / 40**

✓ - **0 pts** Correct

- **10 pts** Patm is wrong
- **5 pts** unit conversion
- **15 pts** P_{gage} is wrong or not calculated
- **15 pts** P absolute is wrong or not calculated
- **40 pts** Not done or incorrect

QUESTION 2

Question 2 60 pts

2.1 a **25 / 30**

- **0 pts** Correct
- **5 pts** system not identified
- **5 pts** engineering model is missing
- **10 pts** PV diagram is wrong
- **15 pts** Work calculation is wrong
- ✓ - **5 pts** unit is not converted
- **30 pts** not done

2.2 b **10 / 30**

- **0 pts** Correct
- ✓ - **20 pts** incorrect or no calculation
- ✓ - **10 pts** PV diagram is wrong
- **0 pts** Click here to replace this description.
- + **10** Point adjustment

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QUIZ 1: 2 problems (10 pts)

Closed book/Closed notes: Show all your work for full credit

1. (6 pts) The average atmospheric pressure on Earth is approximated as a function of altitude by the relation $P_{\text{atm}} = 101.325(1 - 0.02256z)^{5.256}$, where P_{atm} is atmospheric pressure in kPa and z is altitude in km with $z=0$ at sea level. A monometer is used to measure the pressure of a gas in a tank in Denver ($z=1610\text{m}$). The fluid has a density of 850kg/m^3 and the monometer column height is 200cm . Determine the absolute and gage pressure of gas within the tank.

$$P_{\text{abs}} = ?$$

$$P_{\text{atm}} = 101.325(1 - 0.02256z)^{5.256} \text{ kPa}$$

$$P_{\text{gage}} = ?$$

$$P_{\text{atm in Den}} = 101.325(1 - 0.02256(1.61\text{km}))^{5.256}$$

$$= 83.41879 \text{ kPa}$$

$$P_{\text{gage}} = P_{\text{abs}} - P_{\text{atm}}$$

$$P_{\text{abs}} = P_{\text{gage}} + P_{\text{atm}}$$

$$P_{\text{abs}} = \rho gh + P_{\text{atm}}$$

$$= 16.66 \text{ kPa} + 83.41879 \text{ kPa}$$

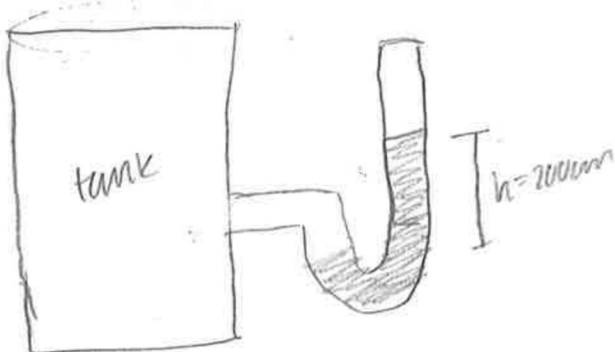
$$= 100.07879 \text{ kPa}$$

$$P_{\text{gage}} = \rho gh$$

$$= 850 \frac{\text{kg}}{\text{m}^3} (9.8 \frac{\text{m}}{\text{s}^2}) (2\text{m})$$

$$= 16660 \text{ N/m}^2$$

$$= 16660 \text{ Pa} \quad \boxed{16.66 \text{ kPa}}$$



$$P_{\text{abs in tank}} = 100.079 \text{ kPa}$$

$$P_{\text{gage in tank}} = 16.66 \text{ kPa}$$

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2. (4 pts) A gas contained within a piston-cylinder assembly undergoes an expansion process for which the relationship between pressure and volume is given by

$PV^n = \text{constant}$

The initial pressure is 2 bar, $V_1 = 2.5 \text{ m}^3$ to $V_2 = 4 \text{ m}^3$. Determine the work for the process in kJ, if

- $n = 0$ and
- $n = 1.5$
- Sketch both processes on a P-V diagram



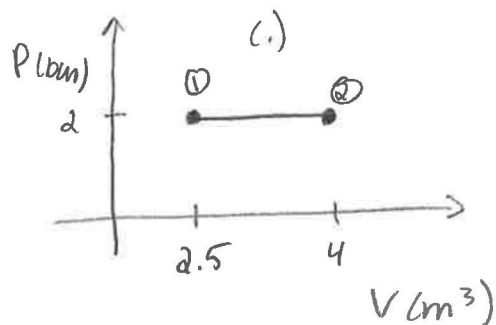
$$P_1 = 2 \text{ bar} \quad P_1 V_1 = P_2 V_2$$

$$V_1 = 2.5 \text{ m}^3$$

$$V_2 = 4 \text{ m}^3$$

~~a. $n=0$~~
 ~~$P_1 V_1^0 = P_2 V_2^0$~~
 ~~2 bar~~

a.) $PV^0 = P$
 $P_1 V_1^0 = P_2 V_2^0$
 $(2 \text{ bar})(2.5 \text{ m}^3)^0 = P_2 (4 \text{ m}^3)^0$
 $P_2 = 2 \text{ bar}$

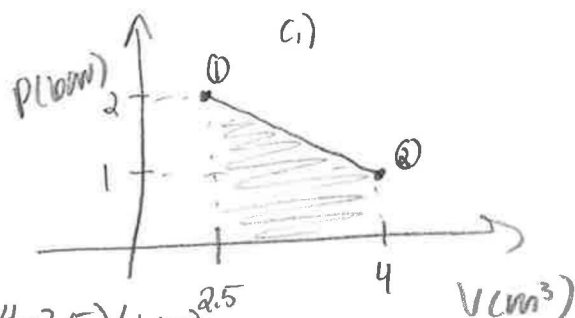


$$W = \int_{V_1}^{V_2} P dV$$

$$= P(V|_{V_1}^{V_2}) = 2 \text{ bar} (4 - 2.5) = 3 \text{ N/m}^2 \cdot \text{m}^3 = 3 \text{ N} \cdot \text{m} = 3 \text{ J}$$

$$= \boxed{1.003 \text{ kJ}}$$

b.) $P_1 V_1^{1.5} = P_2 V_2^{1.5}$
 $(2 \text{ bar})(2.5 \text{ m}^3)^{1.5} = P_2 (4 \text{ m}^3)^{1.5}$
 $P_2 = 0.998 \text{ bar}$



$$W = (0.998 \text{ bar} \cdot (4 - 2.5)) + \frac{1}{2} (4 - 2.5) (1.017)^{2.5}$$

$$= 2.256 \text{ J} = \boxed{0.002256 \text{ kJ}}$$

$$\text{bar} \cdot \text{m}^3 = \frac{\text{N}}{\text{m}^2} \cdot \text{m}^3 = \text{N} \cdot \text{m} = \text{J}$$